

Machine Learning Based Early Detection of Cattle Diseases

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, or material that has been published or written by another person, except where due acknowledgment has been made in the text.

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ABSTRACT

Lumpy Skin Disease (LSD) is a highly contagious viral disease caused by the Lumpy Skin Disease Virus (LSDV), a member of the Capripoxvirus genus, and poses a severe threat to the cattle industry in Sri Lanka. Current diagnostic practices rely predominantly on manual visual inspection by veterinarians, which is subjective, time-consuming, and often results in delayed detection, particularly in rural farming communities with limited veterinary access. Late diagnosis leads to widespread transmission, significant economic losses, and compromised animal welfare.

This research proposes a web-based intelligent detection system that leverages deep learning techniques, specifically YOLOv8 for real-time lesion detection and ResNet50 for classification, to identify Lumpy Skin Disease from cattle skin images. The methodology includes data collection from Sri Lankan farms and veterinary centers, preprocessing of skin images through noise reduction, contrast enhancement, and data augmentation, followed by model training and validation. The system will output a detection result along with a probability percentage indicating the likelihood of LSD presence.

In addition, the proposed system integrates an end-to-end detection-classification pipeline that enables automated lesion localization and disease probability estimation, ensuring efficient and accurate analysis of field-level images. The platform is designed to be farmer friendly, allowing users with minimal technical knowledge to easily upload images and obtain results in real-time. Furthermore, the system incorporates decision-support features by providing risk-based recommendations and generating downloadable reports that can assist veterinary professionals in clinical decision-making.

The anticipated outcome is an automated, scalable, and accessible web platform capable of detecting LSD with high accuracy and providing actionable guidance to farmers, including recommendations to consult a veterinarian when the probability indicates a critical condition. A downloadable report will also be generated for veterinary reference. The system aims to reduce diagnostic delays, minimize economic losses from late detection, enhance disease surveillance capabilities, and improve accessibility to early disease screening across Sri Lankan cattle farms. This project aligns with UN Sustainable Development Goals, notably SDG 2 (Zero Hunger) by enhancing livestock resilience, SDG 3 (Good Health) through improved animal health management, and SDG 12 (Responsible Consumption) by reducing preventable losses.

Keywords: Lumpy Skin Disease, Deep Learning, YOLOv8, ResNet50, Image Processing, Cattle Disease Detection, Web-based Diagnosis

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TABLE OF CONTENTS

DECLARATION	Error! Bookmark not defined.
ABSTRACT	3
ACKNOWLEDGEMENT	4
TABLE OF CONTENTS	5
LIST OF TABLES	7
LIST OF FIGURES	8
LIST OF ABBREVIATIONS	9
LIST OF APPENDICES	10
CHAPTER 1: INTRODUCTION	11
1.1 Background and Context	11
1.2 Literature Review	13
1.3 Research Gap	14
CHAPTER 2: OBJECTIVES.....	16
2.1 Main Objective.....	16
2.2 Specific Objectives	16
CHAPTER 3: METHODOLOGY	17
3.1 Research Approach and Justification.....	17
3.2 Technical Methods and Algorithms.....	17
3.3 Tool and Technology Justification	19
3.4 Data Collection and Ethical Considerations.....	19
3.5 Validation Strategy and Performance Metrics.....	20
3.6 Web-Based Application Development.....	20
CHAPTER 4: HIGH-LEVEL SYSTEM ARCHITECTURE.....	21
4.1 LSD Detection Component Architecture.....	21
4.2 Overall Four-Component System Architecture	21
4.3 Component Interactions and Data Flows.....	22
4.4 Scalability and Security Considerations	23
CHAPTER 5: USER REQUIREMENTS.....	24
5.1 Stakeholder Identification	24
5.2 Functional Requirements	24
5.3 Non-Functional Requirements.....	25

CHAPTER 6: COMMERCIALIZATION PLAN	26
6.1 Target Market and Market Size	26
6.2 Customer Persona	26
6.3 Value Proposition	26
6.4 Revenue Model	26
6.5 Cost Estimation	26
6.6 Competitive Advantage	27
6.7 Intellectual Property Considerations.....	27
CHAPTER 7: BUDGET AND JUSTIFICATION	28
7.1 Budget	28
CHAPTER 8: WORK BREAKDOWN STRUCTURE (WBS)	29
8.1 Tasks and Subtasks	29
8.2 Gantt Chart	29
CHAPTER 9: PROJECT REQUIREMENTS, PERSONNEL, AND FACILITIES	31
9.1 Personnel.....	31
9.2 Facilities and Infrastructure.....	31
REFERENCES	32
APPENDICES	33

LIST OF TABLES

Table 3.1: Tool and Technology Justification

Table 5.1: Functional Requirements

Table 5.2: Non-Functional Requirements

Table 6.1: Commercialization Cost Estimation

Table 7.1: Budget Calculation

LIST OF FIGURES

Figure 1.1: Cattle affected by Lumpy Skin Disease showing characteristic skin nodules

Figure 1.2: Close-up of LSD nodules at various stages of development

Figure 3.1: Workflow of the Proposed Methodology for LSD Detection

Figure 3.2: Example of LSD skin lesions on cattle

Figure 3.3: YOLOv8 and ResNet50 integrated pipeline architecture

Figure 4.1: LSD Detection Component System Architecture

Figure 4.2: Overall Four-Component System Architecture

Figure 4.3: Data Flow Diagram

Figure 8.1: Gantt Chart

LIST OF ABBREVIATIONS

Abbreviation	Description
LSD	Lumpy Skin Disease
LSDV	Lumpy Skin Disease Virus
CNN	Convolutional Neural Network
DL	Deep Learning
ML	Machine Learning
AI	Artificial Intelligence
YOLO	You Only Look Once
mAP	Mean Average Precision
IoU	Intersection over Union
ROI	Region of Interest
DAPH	Department of Animal Production and Health
FMD	Foot and Mouth Disease
CLAHE	Contrast Limited Adaptive Histogram Equalization
SDG	Sustainable Development Goal
PCR	Polymerase Chain Reaction
ELISA	Enzyme-Linked Immunosorbent Assay

LIST OF APPENDICES

Appendix A: Preliminary UI Mockups

Appendix B: Risk Analysis

Appendix C: Data Collection Instruments

Appendix D: Ethical Clearance Application (Placeholder)

CHAPTER 1: INTRODUCTION

1.1 Background and Context

Lumpy Skin Disease (LSD) is a highly contagious viral infection of cattle caused by the Lumpy Skin Disease Virus (LSDV), which belongs to the genus *Capripoxvirus* within the family *Poxviridae* [1]. The disease is characterized by the appearance of firm, raised, circumscribed skin nodules ranging from 2 to 5 centimeters in diameter, fever, enlarged lymph nodes, reduced milk production, and in severe cases, secondary bacterial infections and death [2]. LSD has emerged as a significant transboundary animal disease, with outbreaks reported across Africa, the Middle East, Southeast Asia, and South Asia, including Sri Lanka [3]. The disease is primarily transmitted through arthropod vectors such as mosquitoes, ticks, and biting flies, making it particularly challenging to control in tropical environments [1][2].

In Sri Lanka, the cattle industry plays a vital role in the rural economy, with approximately 1.2 million cattle contributing to dairy production, agricultural labour, and livelihoods for smallholder farmers [4]. The emergence of LSD outbreaks in Sri Lanka has caused significant economic losses due to decreased milk yield, damage to hides, reduced draft power, treatment costs, and trade restrictions [3][5]. The Department of Animal Production and Health (DAPH) has identified LSD as a notifiable disease requiring immediate reporting and containment measures [4]. The major stakeholders affected by this situation are cattle breeders, veterinarians, and related institutions belonging to the agricultural sector, with small-scale cattle breeders being the most impacted.

The effects of delayed disease identification are multifaceted. Economic implications include loss of regular income from lower dairy production and higher treatment costs. Social implications for dairy-reliant rural areas include the threat of loss of livelihood. From a technical perspective, the main issue with existing methods of disease identification is the strong manual observation element. From an ethical viewpoint, delayed detection leads to unnecessary suffering and poor animal welfare. Furthermore, highly contagious diseases like LSD pose the threat of widespread outbreaks across herds [3][4][5].

Current diagnostic methods for LSD rely primarily on clinical examination through visual inspection by veterinarians, who look for characteristic skin nodules and associated symptoms [2]. While effective when performed by experienced professionals, this approach has several critical limitations. Manual visual inspection is subjective and dependent on the expertise of the examining veterinarian [5]. Early-stage nodules, which appear as small, flat elevations before developing into characteristic raised lesions, are particularly difficult to identify and are often overlooked during routine inspections [6]. Furthermore, access to qualified veterinary professionals is limited in many rural farming areas across Sri Lanka, resulting in delayed diagnosis and increased risk of disease transmission within and between herds [3][4].

Laboratory-based confirmatory methods, including Polymerase Chain Reaction (PCR) testing and virus isolation, provide definitive diagnosis but are expensive, time-consuming, and require specialized equipment not available at the farm level [1][7]. Serological tests such as Enzyme-Linked Immunosorbent Assay (ELISA) can detect antibodies but cannot differentiate

between vaccinated and naturally infected animals and have limitations in early detection [2][7].

In recent years, researchers have begun exploring the application of Artificial Intelligence (AI) and Deep Learning (DL) for automated detection of livestock diseases from images. Convolutional Neural Networks (CNNs) have demonstrated remarkable success in image classification and object detection tasks across various domains, including medical imaging and veterinary diagnostics [8]. Specifically, the YOLOv8 (You Only Look Once version 8) architecture has emerged as a state-of-the-art real-time object detection model, capable of accurately identifying and localizing objects in images with high speed and precision [9]. ResNet50, a deep residual network with 50 layers, has been widely used for image classification tasks due to its ability to address the vanishing gradient problem through skip connections [10].

This research is within the purview of CoEAI – Centre of Excellence for Artificial Intelligence under the Artificial Intelligence and Data Science research cluster and falls within the Agriculture and Livestock industry vertical. The project addresses the United Nations Sustainable Development Goals, notably SDG 2 by enhancing the resilience of dairy farming, SDG 3 through enhanced practices in animal health management, and SDG 12 by reducing preventable losses to support sustainable practices in animal livestock.



Figure 1.1: Cattle affected by Lumpy Skin Disease showing characteristic skin nodules



Figure 1.2: Close-up of LSD nodules at various stages of development

1.2 Literature Review

This section reviews recent scholarly research on the application of machine learning and deep learning techniques for livestock disease detection, with a specific focus on skin disease identification in cattle.

Tuppurainen and Oura (2012) provided a comprehensive review of Lumpy Skin Disease as an emerging transboundary threat, documenting its epidemiology, transmission mechanisms, and impact on cattle populations across affected regions. Their work established the foundation for understanding LSD's global significance and the urgent need for improved detection methods [1].

Babiuk et al. (2008) examined capripoxviruses as emerging worldwide threats to livestock, detailing the pathogenesis and diagnostic challenges associated with LSD and related diseases. They highlighted the limitations of conventional laboratory-based diagnostic approaches, particularly their inaccessibility in resource-limited farming contexts [5].

Alharan et al. (2021) applied Convolutional Neural Network (CNN)-based approaches to classify cattle skin diseases from images, achieving accuracy rates above 85%. Their study demonstrated the feasibility of using deep learning for automated skin disease identification in cattle, though the work focused on general skin disease classification rather than LSD-specific detection [8].

Kumar et al. (2022) demonstrated that deep learning models could effectively detect skin lesions in cattle using transfer learning with pre-trained architectures like VGG16 and InceptionV3. Their research showed promising results for lesion detection but did not target LSD specifically, nor did they develop a farmer-accessible detection system [6].

Jocher et al. (2023) developed and released the YOLOv8 architecture by Ultralytics, representing the latest advancement in real-time object detection. YOLOv8 demonstrated state-of-the-art performance across multiple benchmarks with improved accuracy and speed compared to previous YOLO versions, making it suitable for field-deployable livestock disease detection applications [9].

He et al. (2016) introduced ResNet50, a deep residual learning framework that addressed the vanishing gradient problem through skip connections, enabling effective training of very deep networks. ResNet50 has since become a standard architecture for image classification tasks and has been widely adopted in medical and veterinary imaging applications [10].

Calistri et al. (2019) published an epidemiological report on Lumpy Skin Disease through the European Food Safety Authority (EFSA), documenting the disease's spread patterns and emphasizing the importance of early detection for effective containment. Their findings underscored the need for rapid, accessible diagnostic tools [7].

The Department of Animal Production and Health (DAPH), Sri Lanka maintains records of livestock disease outbreaks and manages disease control programs across the country. Their reports indicate that LSD is a notifiable disease in Sri Lanka, and current control

measures rely primarily on vaccination campaigns and clinical surveillance by veterinary officers [3][4].

Murphy et al. (1999) provided foundational veterinary virology knowledge on capripoxviruses, including detailed descriptions of LSD pathology and clinical presentation that inform the morphological features targeted by modern detection models [11].

The Center for Food Security and Public Health, Iowa State University published comprehensive factsheets on Lumpy Skin Disease covering clinical signs, differential diagnosis, and prevention strategies. These resources highlight the visual characteristics of LSD nodules that distinguish them from other skin conditions, which are critical for training image-based detection models [12].

Despite these advances, the reviewed literature reveals that no study has specifically developed a deep learning-based detection system for LSD using YOLOv8 and ResNet50, trained on Sri Lankan cattle data, and deployed as a farmer-accessible web platform with probability-based output and decision support.

1.3 Research Gap

Despite advances in AI-assisted livestock disease diagnostics, several critical gaps remain, particularly with regard to Lumpy Skin Disease detection in Sri Lanka:

Absence of LSD-Specific Deep Learning Models. Most existing AI and ML approaches for cattle disease detection focus on general skin disease classification or target other diseases such as Foot and Mouth Disease and mastitis [6][8]. There is a notable absence of deep learning models specifically trained and optimized for detecting LSD skin nodules, which have unique morphological characteristics that distinguish them from other skin conditions such as dermatophilosis, insect bites, and allergic reactions [1][2].

Lack of Sri Lankan-Specific Data and Models. Existing research models are predominantly trained on datasets collected from other geographical regions, such as Africa and the Middle East [5][6]. Cattle breeds, environmental conditions, skin pigmentation, and farming practices in Sri Lanka differ significantly from these regions, potentially affecting model performance when applied locally [3][4]. No AI-based LSD detection model has been developed using data collected from Sri Lankan farms.

Limited Focus on Early-Stage Detection. Most current diagnostic approaches, both manual and AI-assisted, focus on identifying LSD at advanced stages when nodules are fully developed and clearly visible [2][5]. Early-stage detection, when nodules are small, flat, and easily confused with insect bites or minor skin irritations, remains a significant challenge [6].

Absence of Farmer-Accessible Diagnostic Tools. While some research prototypes for livestock disease detection exist, they are typically designed for laboratory or research environments and are not accessible to farmers [5][8]. There is no existing web-based system that allows Sri Lankan cattle farmers to upload skin images and receive immediate, actionable diagnostic feedback regarding LSD.

No Integration of Detection with Decision Support. Current AI-based detection approaches typically provide binary classification without offering probability-based assessments or actionable guidance [6][8]. A system that provides probability percentages along with context-specific recommendations would be far more useful in practical farming contexts.

In conclusion, there is a clear need for a Sri Lanka-specific, deep learning-based LSD detection system that uses locally collected data, provides probability-based assessments, and delivers actionable guidance to farmers through an accessible web platform.

CHAPTER 2: OBJECTIVES

2.1 Main Objective

To design, develop, and implement a deep learning-based detection system component for Lumpy Skin Disease (LSD) in Sri Lankan cattle that uses YOLOv8 for skin lesion detection and ResNet50 for classification from cattle skin images, providing accurate detection results with probability percentages, actionable guidance for farmers, and a practical, user-friendly web-based platform for early disease screening.

2.2 Specific Objectives

- Collect and preprocess a comprehensive dataset of cattle skin images from Sri Lankan farms and veterinary centers, ensuring representation of LSD-positive cases, other skin conditions, and healthy cattle for robust model training.
- Develop a YOLOv8-based object detection model to identify and localize LSD skin nodules in cattle images with high precision, using transfer learning to optimize performance with locally collected data.
- Develop a ResNet50-based classification model to determine the probability of Lumpy Skin Disease from detected lesion regions, outputting a probability percentage rather than binary classification.
- Design and implement an integrated detection-classification pipeline combining YOLOv8 and ResNet50 that processes uploaded images through sequential detection and classification stages to produce an overall LSD probability score.
- Validate the system using real-world clinical data from Sri Lankan cattle, comparing model predictions against veterinary diagnoses using metrics including mAP, accuracy, precision, recall, and F1-score.
- Develop a farmer-friendly web-based platform that enables image upload, displays detection results with probability percentages and visual bounding boxes, provides actionable guidance based on risk level, and generates downloadable veterinary reports.

CHAPTER 3: METHODOLOGY

3.1 Research Approach and Justification

This research follows an Experimental Research approach combined with Design Science methodology. The experimental component involves training and evaluating deep learning models (YOLOv8 and ResNet50) on cattle skin images under controlled conditions with measurable performance metrics. The design science component involves the iterative design and development of the web-based detection platform as a technological artifact, evaluated through both technical performance and user feedback.

This combined approach is justified because the project requires both rigorous model evaluation (accuracy, precision, recall) and practical system design (usability, accessibility). The experimental methodology ensures the detection models are scientifically validated, while the design science approach ensures the final platform meets the practical needs of Sri Lankan cattle farmers.

3.2 Technical Methods and Algorithms

3.2.1 Image Preprocessing Pipeline

The image preprocessing pipeline prepares field-captured cattle skin images for model input through the following steps:

- **Noise Reduction:** Gaussian blur and median filtering to remove artifacts from images captured in outdoor farm environments.
- **Contrast Enhancement:** Histogram equalization and CLAHE (Contrast Limited Adaptive Histogram Equalization) to improve visibility of skin nodules against varying skin tones and lighting conditions.
- **Image Resizing:** Normalization to standard dimensions required by YOLOv8 (640×640) and ResNet50 (224×224) input layers.
- **Data Augmentation:** Rotation (0°, 90°, 180°, 270°), horizontal and vertical flipping, random cropping, brightness/contrast adjustments, and scaling to increase dataset diversity and reduce overfitting.
- **Annotation:** Bounding box annotation using LabelImg/Roboflow for YOLOv8 training; image-level labels (LSD-positive/negative) for ResNet50 classification training.

3.2.2 YOLOv8 Lesion Detection Model

YOLOv8 (You Only Look Once version 8) by Ultralytics will be used as the primary object detection model to identify and localize potential LSD skin nodules in cattle images [9]. YOLOv8 is selected for its state-of-the-art balance of detection accuracy and inference speed, making it suitable for real-time applications. Transfer learning will be applied using pre-trained YOLOv8 weights (trained on COCO dataset) to accelerate convergence and improve performance with the limited locally collected dataset. The model will output bounding boxes around detected nodule regions with associated confidence scores.

3.2.3 ResNet50 Classification Model

ResNet50 (Residual Network with 50 layers) will classify detected skin regions to determine the probability of LSD [10]. ResNet50 is selected for its proven effectiveness in image classification tasks and its ability to train very deep networks through skip connections that address the vanishing gradient problem. Transfer learning with ImageNet pre-trained weights will be used, with the final fully connected layers replaced and fine-tuned for binary classification (LSD-positive vs. LSD-negative). The softmax output layer will provide a probability percentage indicating the likelihood of LSD.

3.2.4 Integrated Detection-Classification Pipeline

The YOLOv8 and ResNet50 models will be combined into a sequential pipeline:

- Step 1: The input image is processed by YOLOv8 to detect potential lesion regions, producing bounding boxes with confidence scores.
- Step 2: Detected regions are cropped from the original image and resized for ResNet50 input.
- Step 3: ResNet50 classifies each detected region, determining the probability of LSD.
- Step 4: The final output combines detection confidence and classification probability to produce an overall LSD probability percentage.
- Step 5: If no lesions are detected by YOLOv8, the system reports “No suspicious lesions detected” with a low probability score.

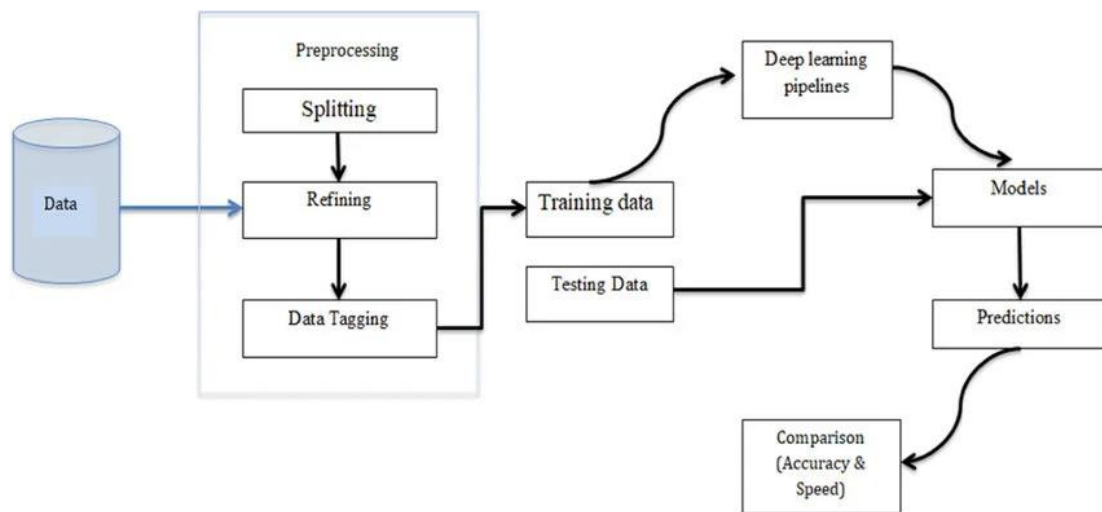


Figure 3.3: YOLOv8 and ResNet50 integrated pipeline architecture

3.3 Tool and Technology Justification

Tool/Technology	Purpose	Justification
Python 3.9+	Programming language	Industry standard for ML/DL with extensive library support
TensorFlow/PyTorch	Deep learning framework	Mature frameworks with strong community and GPU support
Ultralytics YOLOv8	Object detection	State-of-the-art real-time detection with excellent accuracy-speed tradeoff
ResNet50	Image classification	Proven architecture for classification with effective transfer learning
OpenCV	Image processing	Comprehensive library for image preprocessing and augmentation
Scikit-image	Image processing	Additional image analysis and segmentation capabilities
LabelImg/Roboflow	Data annotation	Standard tools for bounding box annotation for object detection
Flask/Django	Backend web framework	Lightweight and scalable frameworks for serving ML models via REST API
React/Angular	Frontend framework	Modern responsive UI frameworks for web application development
MongoDB/PostgreSQL	Database	Flexible storage for image metadata, results, and user data
Google Colab Pro/AWS	Cloud computing	GPU-accelerated model training when local hardware is insufficient

3.4 Data Collection and Ethical Considerations

3.4.1 Primary Data Sources

- Cattle skin images captured from Sri Lankan farms and veterinary centers using standard smartphone cameras under natural lighting conditions.
- Images will include: LSD-confirmed cases (nodules at various stages), cattle with other skin conditions (dermatophilosis, insect bites, allergic reactions), and healthy cattle skin as negative samples.
- Ground truth labels established through clinical diagnosis by qualified veterinarians.
- Multiple angles and distances captured for each case to ensure diversity.

3.4.2 Secondary Data Sources

- Reports and disease surveillance data from the Department of Animal Production and Health (DAPH), Sri Lanka.
- Published literature on LSD clinical presentation and morphological features.

3.4.3 Ethical Considerations

- Ethical clearance is currently being obtained from the relevant institutional review committee at SLIIT.
- Consent from farm owners will be secured before any data collection.
- No invasive procedures will be performed on animals during image capture.
- All data will be anonymized, removing identifiable information about farms and owners.
- Data will be stored securely with access restricted to authorized project members only.

3.5 Validation Strategy and Performance Metrics

The system will be validated using the following strategy:

- Dataset Split: Training (70%), Validation (15%), Testing (15%) with balanced class representation.
- YOLOv8 Metrics: Mean Average Precision (mAP@0.5), Precision, Recall, and Intersection over Union (IoU).
- ResNet50 Metrics: Accuracy, Precision, Recall, F1-Score, and AUC-ROC curve.
- Integrated Pipeline Metrics: Overall detection accuracy, false positive rate, false negative rate, and end-to-end processing time.
- K-fold cross-validation to ensure model generalizability across different data subsets.
- Comparison of system predictions against veterinary clinical diagnoses to assess real-world reliability.

3.6 Web-Based Application Development

The web-based platform will be developed with the following features:

- Image Upload Interface: Accepts cattle skin images in JPEG/PNG formats through a responsive web interface.
- Automated Detection Pipeline: Processes uploaded images through the YOLOv8 + ResNet50 pipeline.
- Visual Output: Displays the uploaded image with bounding boxes around detected lesions and confidence scores.
- Probability Display: Shows the overall LSD probability percentage prominently.
- Actionable Guidance based on probability thresholds:
 - Low probability (0–30%): “Low risk. Continue monitoring. Maintain regular health checks.”
 - Medium probability (30–70%): “Moderate risk detected. Isolate the animal and monitor closely. Consider consulting a veterinarian.”
 - High probability (70–100%): “High risk of LSD detected. Immediate veterinary consultation is strongly advised. Isolate the animal from the herd.”
- Report Generation: Downloadable PDF report containing image, results, probability, and recommendations for veterinary reference.
- No veterinary doctor login is required. The system is designed for direct use by farmers.

CHAPTER 4: HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 LSD Detection Component Architecture

The LSD Detection component follows a sequential pipeline architecture consisting of four layers:

- **User Input Layer:** The farmer uploads a cattle skin image through the web interface.
- **Preprocessing Layer:** The image undergoes noise reduction, contrast enhancement, resizing, and normalization.
- **Detection & Classification Layer:** YOLOv8 detects potential lesion regions, which are then classified by ResNet50 to determine LSD probability.
- **Output Layer:** The system displays the probability percentage, visual output with bounding boxes, actionable guidance, and a downloadable report.

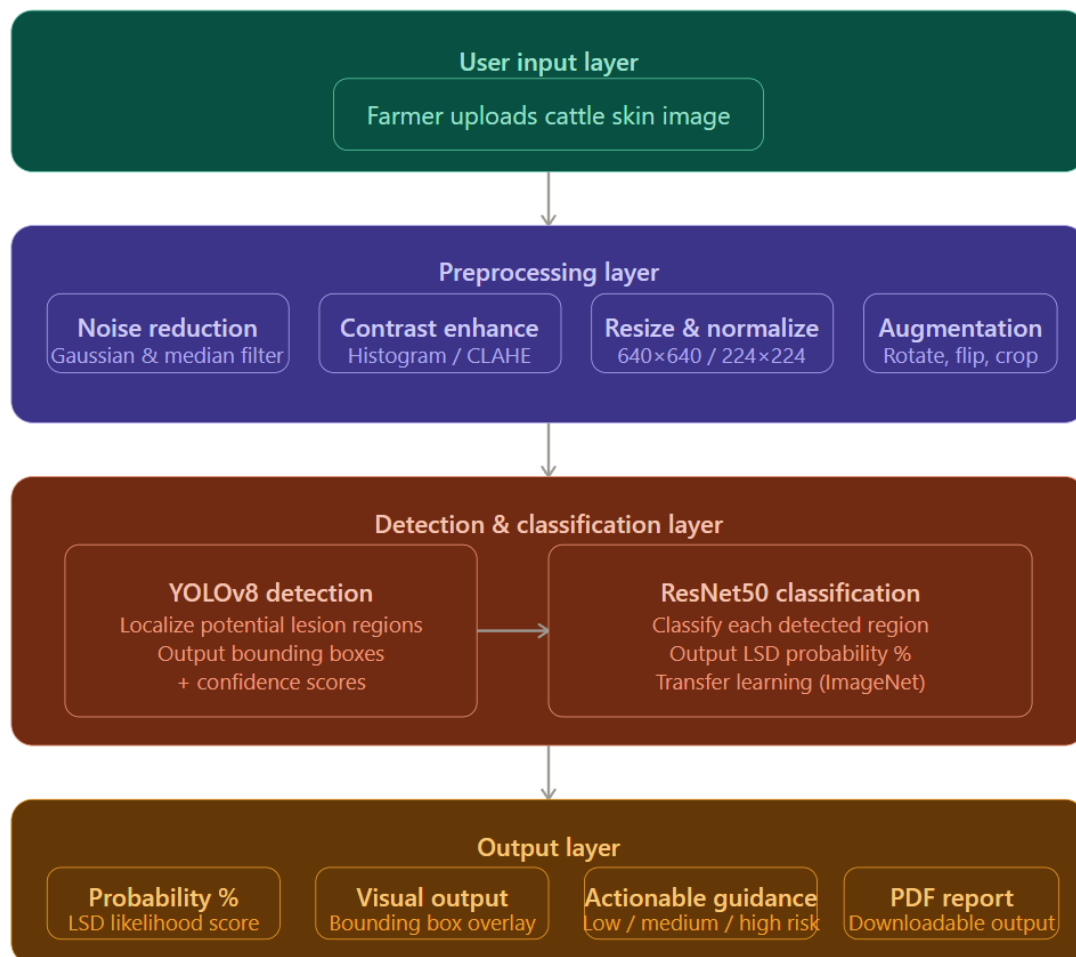


Figure 4.1: LSD Detection Component System Architecture

4.2 Overall Four-Component System Architecture

The LSD Detection module is one of four components in the broader “Machine Learning Based Early Detection of Cattle Diseases” system:

- Component 1 – Mastitis Detection (Perera L.C.C, IT22153036): Uses udder images and health data with CNN-based feature extraction and ML classification.
- Component 2 – Foot and Mouth Disease Detection (Sudasingha D.D, IT22129512): Uses mouth and hoof images with CNN-based lesion detection and classification.
- Component 3 – Lumpy Skin Disease Detection (Manathunga M.A.A.S, IT22282422): Uses cattle skin images with YOLOv8 detection and ResNet50 classification.
- Component 4 – Respiratory Disease Detection (Udawaththa M.P.A.M, IT22221725): Uses physiological and behavioral data with ML classification.

All four components share a unified web-based platform with a common user interface, authentication system, and database infrastructure.

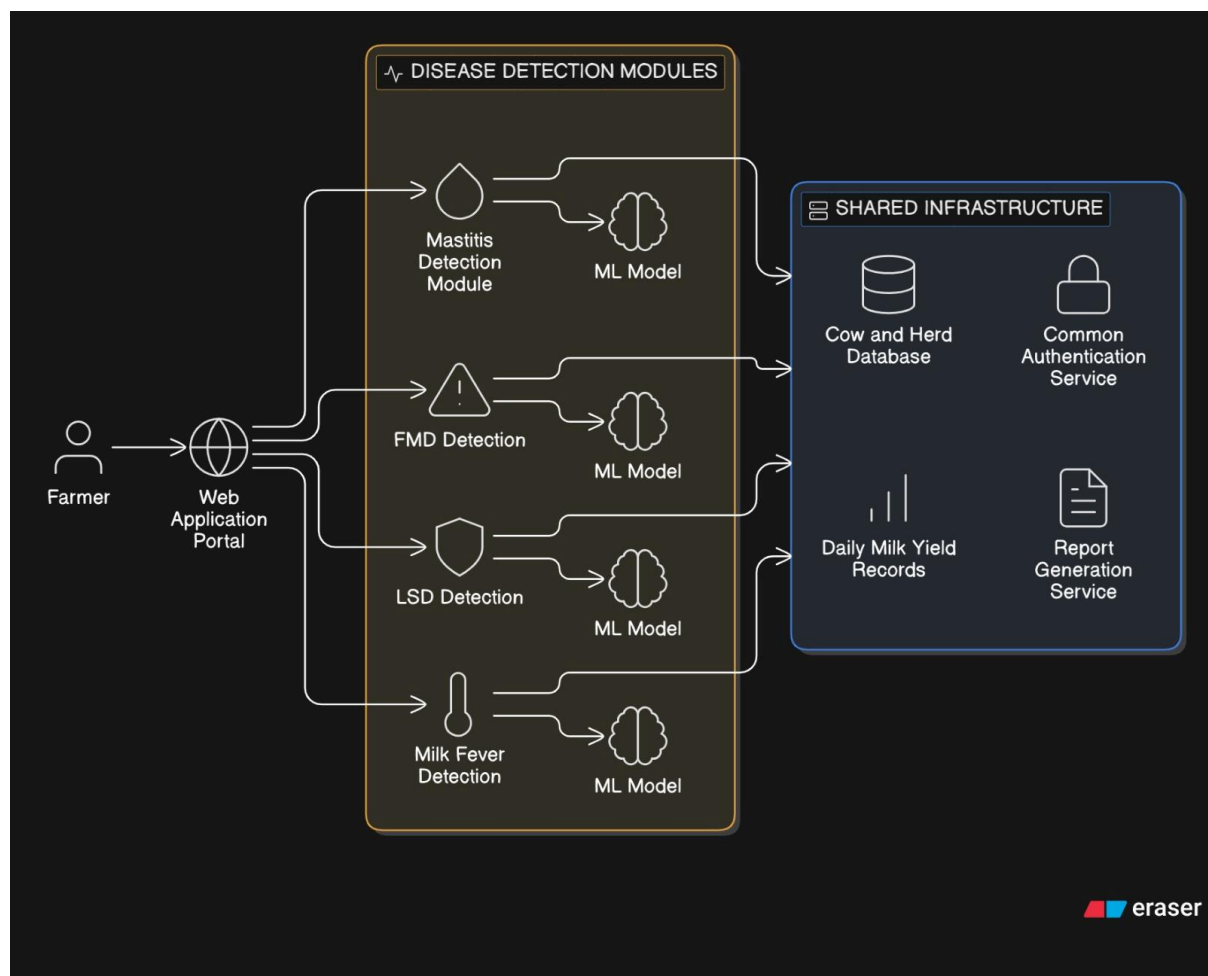


Figure 4.2: Overall Four-Component System Architecture

4.3 Component Interactions and Data Flows

The data flow for the LSD Detection component proceeds as follows:

- The farmer accesses the web platform and selects the LSD Detection module.
- The farmer uploads a cattle skin image through the interface.
- The image is sent to the backend server via REST API.

- The backend preprocesses the image and passes it through the YOLOv8 detection model.
- Detected regions are extracted and classified by ResNet50.
- The integrated decision engine calculates the overall probability and generates guidance.
- Results (probability, visual output, guidance, report) are returned to the frontend and displayed.
- The detection record is stored in the database for future reference.

4.4 Scalability and Security Considerations

Scalability:

- The system uses a modular architecture allowing independent scaling of each disease detection component.
- The backend API can be horizontally scaled to handle increasing numbers of concurrent users.
- Model inference can be offloaded to GPU-enabled cloud instances during peak usage.
- The database design supports efficient storage and retrieval of growing image datasets and detection records.

Security:

- No personally identifiable information about farmers is collected or stored.
- All uploaded images and reports are stored securely with access controls.
- Communication between frontend and backend uses HTTPS encryption.
- User sessions are managed securely to prevent unauthorized access.

CHAPTER 5: USER REQUIREMENTS

5.1 Stakeholder Identification

- Primary Users – Cattle Farmers: Small-scale and medium-scale cattle farmers in Sri Lanka who need to screen their cattle for LSD without immediate veterinary access.
- Secondary Users – Veterinary Officers: Veterinarians who receive generated reports from farmers and use them to inform their clinical assessments.
- Tertiary Users – Agricultural Authorities: Institutions such as DAPH that may use aggregated data for disease surveillance and outbreak monitoring.

5.2 Functional Requirements

ID	Requirement	Description
FR-01	Image Upload	The system shall accept cattle skin images in JPEG and PNG formats through a web interface.
FR-02	Image Preprocessing	The system shall automatically preprocess uploaded images (resize, enhance, denoise) for model input.
FR-03	Lesion Detection	The system shall use YOLOv8 to detect and localize potential skin nodules in uploaded images.
FR-04	Disease Classification	The system shall use ResNet50 to classify detected regions and output an LSD probability percentage.
FR-05	Visual Output	The system shall display the uploaded image with bounding boxes around detected lesions.
FR-06	Guidance Generation	The system shall provide actionable recommendations based on probability thresholds (low/medium/high).
FR-07	Report Generation	The system shall generate a downloadable PDF report with image, results, and recommendations.
FR-08	Detection History	The system shall store previous detection results for user reference.
FR-09	Data Storage	The system shall store anonymized data securely for model improvement.

5.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Accuracy	The detection model shall achieve minimum mAP of 80% and classification accuracy above 85%.
NFR-02	Performance	Detection results shall be generated within 10–15 seconds after image upload.
NFR-03	Usability	The interface shall be simple and intuitive for farmers with limited technical expertise.
NFR-04	Security	All uploaded data shall be stored securely with no personally identifiable information collected.
NFR-05	Reliability	The system shall provide consistent results across varying image quality and lighting conditions.
NFR-06	Availability	The web platform shall be available 24/7 with minimal downtime.
NFR-07	Scalability	The architecture shall support increasing numbers of users and image uploads.
NFR-08	Maintainability	The codebase shall be well-documented and modular for easy updates.

CHAPTER 6: COMMERCIALIZATION PLAN

6.1 Target Market and Market Size

The primary target market is Sri Lanka's dairy farming sector, which comprises approximately 300,000 smallholder dairy farmers and over 1.2 million cattle across the country [4]. Secondary markets include veterinary clinics, government agricultural extension services, and dairy cooperatives. The broader Southeast Asian livestock industry represents a potential expansion market, as LSD has emerged as a significant transboundary disease across the region [1][3].

6.2 Customer Persona

The primary customer persona is a small-scale Sri Lankan cattle farmer, typically managing 5–20 cattle, located in a rural area with limited access to veterinary services. This farmer is familiar with basic smartphone usage but has no technical or veterinary expertise. They need a quick, affordable way to check whether their cattle's skin condition might be LSD, and they need clear guidance on what to do next.

6.3 Value Proposition

- **Early Detection:** Identifies potential LSD cases before clinical signs become severe, enabling timely intervention.
- **Accessibility:** Provides diagnostic screening capability to farmers in areas without immediate veterinary access.
- **Cost-Effective:** Eliminates the need for expensive laboratory tests for initial screening.
- **Actionable Guidance:** Goes beyond detection to advise farmers on specific actions, including when to seek veterinary help.
- **Report Generation:** Creates professional reports that farmers can present to veterinarians, improving communication and care quality.

6.4 Revenue Model

- **Freemium Model:** Basic detection (limited scans per month) available for free; premium subscription for unlimited scans, detailed analytics, and priority support.
- **Institutional Licensing:** Government agencies and dairy cooperatives can license the platform for their network of farmers.
- **Data Insights:** Anonymized, aggregated disease prevalence data can be offered to agricultural research institutions and policymakers (with appropriate consent and privacy safeguards).

6.5 Cost Estimation

Cost Category	Annual Estimate (LKR)
Cloud Hosting & GPU Inference	120,000
Domain & SSL Certificate	5,000
Maintenance & Updates	100,000
Marketing & Outreach	50,000
Total Annual Operating Cost	275,000

6.6 Competitive Advantage

- **First-Mover:** No existing LSD-specific detection system for Sri Lankan cattle.
- **Locally Trained:** Models trained on Sri Lankan data for superior local accuracy.
- **Farmer-Centric Design:** Interface designed for non-technical users, unlike research prototypes.
- **Integrated Ecosystem:** Part of a four-disease detection platform providing comprehensive cattle health screening.

6.7 Intellectual Property Considerations

The trained deep learning models, integrated pipeline architecture, and web platform design constitute the primary intellectual property of this project. The project will use open-source frameworks (TensorFlow/PyTorch, YOLOv8, React) with permissive licenses. The unique value lies in the locally trained model weights and the integrated system design. Potential IP protection strategies include copyright protection for the software and trade secret protection for the trained model weights.

CHAPTER 7: BUDGET AND JUSTIFICATION

7.1 Budget

Item	Description	Estimated Cost (LKR)
Cloud GPU/Computing	Google Colab Pro+ or AWS for YOLOv8 and ResNet50 model training	15,000
Web Deployment	Domain name registration and hosting for the detection platform	8,000
Data Collection & Logistics	Travel to farms and vet centers, printing consent forms and ethical clearance documents	7,000
Miscellaneous	Stationery, internet, contingency expenses	5,000
Total		35,000

Budget Justification:

- Cloud GPU/Computing: Training YOLOv8 and ResNet50 models requires significant GPU resources. Free tier services have limited GPU availability and session durations, making a minimal allocation for cloud GPU services necessary during intensive training phases.
- Web Deployment: A domain name and basic hosting are required to deploy and test the web-based detection platform in real-world conditions.
- Data Collection and Logistics: This covers transportation costs for visiting farms and veterinary centers to collect cattle skin images, as well as printing consent forms, ethical clearance documentation, and data collection instruments.
- Miscellaneous: Includes minor expenses such as stationery, internet data charges, and a small contingency reserve for unforeseen costs.

CHAPTER 8: WORK BREAKDOWN STRUCTURE (WBS)

8.1 Tasks and Subtasks

Task 1: Data Collection

- 1.1: Identify and establish partnerships with Sri Lankan farms and veterinary centers
- 1.2: Obtain ethical clearance from institutional review committees
- 1.3: Capture cattle skin images (LSD-positive, other conditions, healthy) following guidelines
- 1.4: Obtain veterinary clinical diagnoses for ground truth labelling

Task 2: Data Preprocessing

- 2.1: Apply noise reduction and contrast enhancement to collected images
- 2.2: Resize and normalize images for model input requirements
- 2.3: Annotate images with bounding boxes for YOLOv8 training
- 2.4: Label images for ResNet50 classification training
- 2.5: Implement data augmentation techniques
- 2.6: Split dataset into training, validation, and testing sets

Task 3: Model Development

- 3.1: Train YOLOv8 model for LSD lesion detection using transfer learning
- 3.2: Train ResNet50 model for LSD classification using transfer learning
- 3.3: Design and implement the integrated detection-classification pipeline
- 3.4: Implement probability calculation and threshold-based risk assessment logic

Task 4: Validation

- 4.1: Evaluate YOLOv8 performance (mAP, Precision, Recall, IoU)
- 4.2: Evaluate ResNet50 performance (Accuracy, Precision, Recall, F1-Score, AUC-ROC)
- 4.3: Evaluate integrated pipeline against veterinary diagnoses
- 4.4: Perform k-fold cross-validation

Task 5: Web Platform Development

- 5.1: Develop backend API using Flask/Django
- 5.2: Develop frontend interface using React/Angular
- 5.3: Implement visual output with bounding box overlays
- 5.4: Implement guidance generation and report download functionality

Task 6: Field Evaluation & Integration

- 6.1: Conduct pilot testing with farmers and veterinary officers
- 6.2: Collect usability feedback and refine the system
- 6.3: Integrate with the broader four-component cattle disease detection platform

8.2 Gantt Chart

Task	Month 1-2	Month 3-4	Month 5-6	Month 7-8
Data Collection				
Data Preprocessing				
Model Development				
Validation				
Web Platform				
Field Evaluation				
Report Writing				

Figure 8.1: Insert Gantt Chart here – Semester-wise breakdown with milestones

CHAPTER 9: PROJECT REQUIREMENTS, PERSONNEL, AND FACILITIES

9.1 Personnel

The project is being conducted by a group of four final-year undergraduate students specializing in software engineering at the Sri Lanka Institute of Information Technology. Each member develops a unique component of the overall system:

Member 1 – Perera L.C.C (IT22153036) – Mastitis Detection: Responsible for developing the Mastitis detection component using udder images and health data.

Member 2 – Sudasingha D.D (IT22129512) – FMD Detection: Responsible for developing the Foot and Mouth Disease detection component using mouth and hoof images.

Member 3 – Manathunga M.A.A.S (IT22282422) – LSD Detection: Responsible for developing the Lumpy Skin Disease detection component using cattle skin images with YOLOv8 and ResNet50.

Member 4 – Udawaththa M.P.A.M (IT22221725) – Respiratory Disease Detection: Responsible for developing the Respiratory Disease detection component using physiological and behavioral data.

Supervisor: Dr. Prasanna Sumathipala

Co-supervisor: Mrs. Thilini Jayalath

9.2 Facilities and Infrastructure

- University laboratories equipped with high-performance computers and internet access for model training and development.
- Software resources: TensorFlow, PyTorch, Ultralytics YOLOv8, OpenCV, and Scikit-learn (open-source frameworks).
- Web development tools: React.js, Node.js, Flask/Django, and database management systems.
- Data sources: Partnerships with farms and veterinary centers across Sri Lanka for image collection.
- Cloud computing: Google Colab Pro or AWS for GPU-accelerated model training.
- Annotation tools: LabelImg and Roboflow for bounding box annotation.
- Ethical and organizational support from university supervisors for ethical clearance and farm partnerships.

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APPENDICES

Appendix A: Preliminary UI Mockups

[Insert preliminary UI mockup screenshots or wireframes here]

Appendix B: Risk Analysis

Risk	Impact	Likelihood	Mitigation Strategy
Insufficient training data	High	Medium	Supplement with data augmentation; explore publicly available datasets
Low model accuracy	High	Medium	Iterative hyperparameter tuning; experiment with multiple architectures
Ethical clearance delays	Medium	Medium	Begin application early; maintain communication with review committee
Poor image quality from farms	Medium	High	Provide image capture guidelines; robust preprocessing pipeline
Hardware limitations for training	Medium	Low	Utilize cloud GPU services (Colab Pro, AWS)
User adoption challenges	Medium	Medium	Design simple UI; conduct usability testing with farmers

Appendix C: Data Collection Instruments

Image Capture Protocol:

- Use smartphone camera with minimum 8MP resolution
- Capture images under natural daylight when possible
- Include full view and close-up of affected skin areas
- Record animal ID, date, location, and veterinary assessment for each image

Appendix D: Ethical Clearance Application (Placeholder)

[Ethical clearance application is currently in progress. The completed form will be included upon approval.]